

Lecture 1 : Introduction and Basic Calculus

Piotr Zwiernik

19BE02: Calculus and Linear Algebra



Course structure

Instructor: Piotr Zwiernik (piotr.zwiernik@bse.eu).

Five lectures: today, Fri, Mon, Tue, Wed.

Each day: 4 hours of lectures, 1 hour of tutorials.

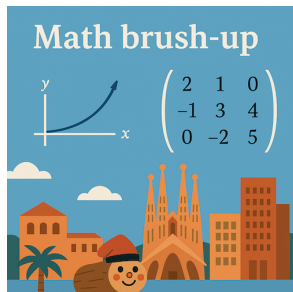
Lectures: 11-1pm and 2-4pm.

Tutorials: 4-5pm.

TA: Francisco Calvillo (francisco.calvillo@bse.eu).

Grading: This course is not graded. But there will be an exam on Friday, Sept 19th, 10-11am.

Course website: <https://pzwiernik.github.io/math-brushup/>.



Goal of the course

Review core tools of calculus that power modern economics, finance, and data-driven policy.

These tools are **crucial** in courses on micro/macro, econometrics, finance, and machine learning.

Warm up before the Master starts: sharpen intuition, not just technique.

Main references:

1. *Mathematics for Economists* by C. P. Simon and L. Blume
 2. *Mathematics of Economics and Business* by F. Werner and Y. N. Sotskov
- (any other standard textbook should be fine)

Course content (math tapas)

**Sequences
and series**



**Functions
of one variable**



Differentiation



Vectors



Integration



**Matrices and
determinants**



**Systems of
linear equations**



**Eigenvalue
problems and
quadratic forms**



Cultivate a Critical Mindset

Technical knowledge requires effort.

- **If a step feels “magical”, stop.** Ask *why* it works, not just *how*.
- **Challenge every answer — especially given by LLMs.**
Large-language models can sound confident yet be wrong; verify with first principles or a textbook.
- **Use multiple lenses.**
Check special cases, sketch a graph, test numerically, or search for a counterexample.

Welcome to Our Class Q&A Forum!

Purpose: A Space for Collaborative Learning

This forum is our dedicated space for all questions related to course content. It's a place for you to ask for help and to help your classmates.

How to Access

1. Go to the **Classwork** tab.
2. Find the topic "**Student Q&A Forum**".
3. Click on the question thread to view discussions or ask your own!

Be Active!

- **Ask:** Don't hesitate to post a question.
- **Answer:** If you know the answer to a classmate's question, please reply!

I will check the forum 2-3 times a day.

Sequences

Sequences

A **sequence** is an ordered list of real numbers $(a_1, a_2, \dots)$ indexed by $n \in \mathbb{N}$.

Notation: $\{a_n\}_{n=1}^{\infty}$ where each $a_n \in \mathbb{R}$.

Defined by a **formula** or a **recursion**.

Examples:

1. $a_n = \frac{n-1}{2n+1}$ ($a_1 = 0, a_2 = \frac{1}{5}, a_3 = \frac{2}{7}, \dots$)

2. $a_1 = 2, a_{n+1} = 2a_n - 1$ ($2, 3, 5, 9, \dots$)

3. $a_1 = 1, a_2 = 1, a_{n+1} = a_n + a_{n-1}$ (Fibonacci)

4. **Arithmetic:** $a_n = a_1 + (n-1)d$

5. **Geometric:** $a_n = a_1 r^{n-1}$

Why sequences show up in modern econ/data

Where they appear:

- **User metrics:** daily active users (DAU) over days n .
- **Inflation tracking:** monthly CPI/CPIH index a_n .
- **Training loss:** $a_n = \text{loss after } n \text{ gradient steps}$ (decreasing sequence).
- **Carbon budgets:** remaining allowance after n years.

Read Ch. 2 Werner-Sotskov. Exercises: 2.4, 2.5, 2.7 (Werner-Sotskov)

Small motivating example: Creator economy revenue

Imagine a YouTuber uploading videos one by one. Each video gets a certain number of views:

$$v_1, v_2, v_3, \dots$$

and YouTube pays the creator a fixed rate of c euros per 1,000 views (*RPM*).

The revenue from video k is

$$r_k = c \cdot \frac{v_k}{1000}.$$

The total revenue after n videos is then the sequence

$$S_n = r_1 + r_2 + \dots + r_n.$$

This is also an example of a **partial sum**; more on that in a second.

Small motivating example: EAR vs APR in FinTech

Let $a_0 = P$ be the initial investment. If a platform advertises an *annual percentage rate* (APR) R but applies interest m times per year, then after n years:

$$a_n = P \left(1 + \frac{R}{m} \right)^{mn}.$$

The one-year *effective annual rate* (EAR) is:

$$\frac{a_1 - a_0}{a_0} = \left(1 + \frac{R}{m} \right)^m - 1.$$

(APR is the quoted yearly rate ignoring compounding; APY/EAR accounts for the fact that interest itself earns interest when applied multiple times in a year.)

What is the actual daily interest? (ignoring fineprint)

Revolut

Personal

Business

Kids & Teens

Company

Log in

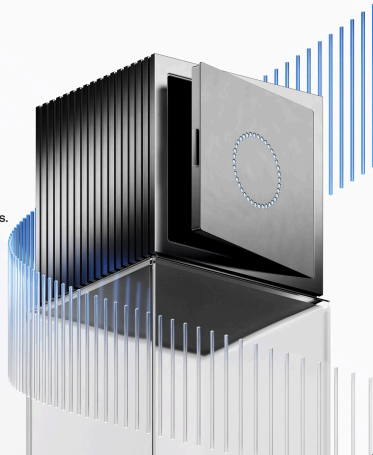
Sign up

Instant Access Savings

SAVINGS, ON DEMAND

Up to 4.5% AER (variable) interest paid daily. No withdrawal fees. No minimums. Instant access anytime.

Get started



Properties of a sequence

A sequence is **increasing** if $a_{n+1} \geq a_n$ (strictly if $>$) for all $n = 1, 2, \dots$

Decreasing if $a_{n+1} \leq a_n$ (strictly if $<$).

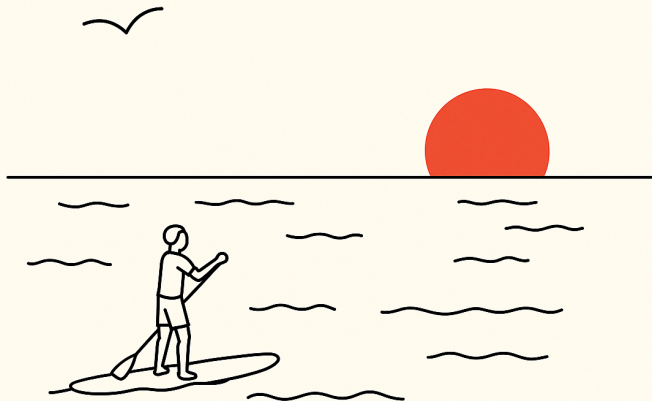
Monotone if increasing or decreasing.

Bounded if $\exists C > 0$ s.t. $|a_n| \leq C$ for all n .

Examples:

1. $a_n = 2(n-1)^2 - n$ is strictly increasing (since $a_{n+1} - a_n = 4n - 3 > 0$)
2. $a_n = 1/n^2$ is bounded ($0 < a_n \leq 1$)
3. **Training loss:** if a learning rate is sensible, $a_n = \text{loss}$ often decreases and is bounded below by 0 $\Rightarrow$ convergent in many practical cases.

Can a strictly increasing sequence be bounded?



Limit of a sequence

Limit of a sequence

A sequence $(a_n)_{n \in \mathbb{N}}$ **converges to zero** if a_n get arbitrarily close to zero.

Also, $a \in \mathbb{R}$ is the **limit** of $(a_n)_{n \in \mathbb{N}}$ if, as $n \rightarrow \infty$, a_n gets arbitrarily close to a .
(equivalently $d_n = |a_n - a|$ converges to zero)

Notation: $\lim_{n \rightarrow \infty} a_n = a$.

A sequence is **convergent** if it has a limit; otherwise it **diverges**.

Examples

1. $a_n = \frac{1}{n^2} \rightarrow 0$
2. $a_n = (-1)^n$ has no limit
3. $a_n = n^2 \rightarrow \infty$
4. $a_n = r^n \rightarrow 0$ if $|r| < 1$; diverges if $r > 1$; no limit if $r \leq -1$.

Formal definition

How do you formalize the statement “gets arbitrarily close”?

Definition: $a = \lim_{n \rightarrow \infty} a_n$

$$\forall \epsilon > 0 \quad \exists N_\epsilon \quad \text{s.t.} \quad \forall n > N_\epsilon : \quad |a_n - a| < \epsilon.$$

Example: $a_n = \frac{1}{n} \rightarrow 0$. For any $\epsilon > 0$, take $N_\epsilon > \frac{1}{\epsilon}$.

Checks:

- $\lim a_n = a$ if and only if $\lim |a_n - a| = 0$.
- If $a_n = a$ for all n , then $\lim a_n = a$.

Theorem(Squeeze theorem): Consider three sequences (a_n) , (b_n) , (c_n) .

If $a_n \leq b_n \leq c_n$ for all $n \in \mathbb{N}$ and $\lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} c_n = L$ then $\lim_{n \rightarrow \infty} b_n = L$.

e.g. $0 < \frac{1}{2n^2-1} \leq \frac{1}{n}$ for $n \in \mathbb{N}$, then $\lim_{n \rightarrow \infty} \frac{1}{2n^2-1} = 0$.

Squeeze theorem



Some limits that matter in practice

Classics

1. $\left(1 + \frac{1}{n}\right)^n \rightarrow e$

2. $\left(1 + \frac{x}{n}\right)^n \rightarrow e^x$ for any real x

Continuous compounding & EAR: If APR is R , then

$$\lim_{m \rightarrow \infty} \left(1 + \frac{R}{m}\right)^m = e^R, \quad \text{EAR} \approx e^R - 1.$$

Example: APR = 8% $\Rightarrow$ EAR $\approx$ 8.329%.

With monthly compounding ($m = 12$) EAR is 8.3%.

With daily compounding ($m = 365$) EAR is 8.328%

Monotone convergence and algebra of limits

Theorem

Every bounded monotone sequence is convergent.

(Just bounded or just monotone is not enough in general)

Example: the sequence $a_1 = 1$, $a_{n+1} = \sqrt{3a_n}$ is bounded by 3 (proof by induction) and strictly increasing (since $\frac{a_{n+1}}{a_n} > 1$), thus convergent.

Algebra of limits

If $\lim a_n = a$ and $\lim b_n = b$, then

$$\lim(a_n + b_n) = a + b, \quad \lim(a_n b_n) = ab. \quad \text{Moreover, } \lim \frac{a_n}{b_n} = \frac{a}{b} \text{ if } b_n, b \neq 0.$$

Series

Partial sums and series

The n th partial sum s_n of $\{a_n\}$ is

$$s_n = \sum_{k=1}^n a_k.$$

Examples

1. Arithmetic sequence $a_n = a_1 + (n-1)d$: $s_n = na_1 + \frac{n(n-1)}{2}d$
2. Geometric sequence $a_n = a_1 r^{n-1}$, $r \neq 1$: $s_n = a_1 \frac{1-r^n}{1-r}$

Use-cases:

- Let *margin* be the profit earned per customer per period and p be the customer retention rate. The total profit from a single customer over n periods is:
 $s_n = \sum_{k=1}^n \text{margin} \cdot p^k$.
- Let E_k be the emissions produced in period k (e.g., in tonnes of CO₂). The total emissions over n periods are: $s_n = \sum_{k=1}^n E_k$, which can be compared to a fixed carbon budget.

Series and convergence

Partial sums also form a sequence. A **series** is the limit (if it exists) of partial sums:

$$\sum_{k=1}^{\infty} a_k := \lim_{n \rightarrow \infty} s_n.$$

Geometric series: converges if and only if $|r| < 1$ with

$$\sum_{k=1}^{\infty} a_1 r^{k-1} = \frac{a_1}{1-r} \quad (\text{we can show this formally}).$$

Customer Lifetime Values: Consider the example on the previous slide. With margin m and monthly retention $p < 1$, discounted at monthly rate d ,

$$\text{CLV} = \sum_{k=1}^{\infty} \frac{m p^k}{(1+d)^k} = \frac{m \frac{p}{1+d}}{1 - \frac{p}{1+d}} = \frac{mp}{1+d-p}.$$

Chapter 2: Functions of one variable

A **one-variable function** assigns a unique $y \in \mathbb{R}$ to each $x \in \mathbb{R}$: $y = f(x)$.

x : independent (exogenous) variable; y : dependent (endogenous) variable.

Examples in econ/data: cost, revenue, demand, utility, probability of purchase vs. price, click-through rate vs. ad spend, etc.

Read Ch. 3 Werner-Sotskov; Simon-Blume 2.1-2.2, 5.1-5.3.

Graph and domain

Definition

The **graph** of f is $\{(x, y) \in \mathbb{R}^2 : y = f(x)\}$.

The **domain** D is the set of x where $f(x)$ is defined; $D \subseteq \mathbb{R}$.

Notation: $f : D \rightarrow \mathbb{R}$.

Example: $f(x) = \frac{1}{x}$ has $D = \mathbb{R} \setminus \{0\}$; $f(x) = \sin(x)$, $D = \mathbb{R}$.

Sometimes we restrict the domain (e.g., prices $x \geq 0$).

Properties of functions

A function $f : D \rightarrow \mathbb{R}$ is **increasing** if $f(x_1) \leq f(x_2)$ for any $x_1 < x_2$ in D (strict if $f(x_1) < f(x_2)$).

Decreasing: reverse inequality.

Bounded: $\exists C > 0$ s.t. $|f(x)| \leq C$ for all $x \in D$.

Examples:

- **Price-demand curve** often decreasing on realistic intervals (then zero).
- **Learning curve** typically decreasing as algorithm progresses.

Convex/concave functions

For $x_1 < x_2$ and $t \in [0, 1]$, the **convex combination** is $tx_1 + (1 - t)x_2$.

A set $D \subset \mathbb{R}$ is **convex** if it contains convex combinations.

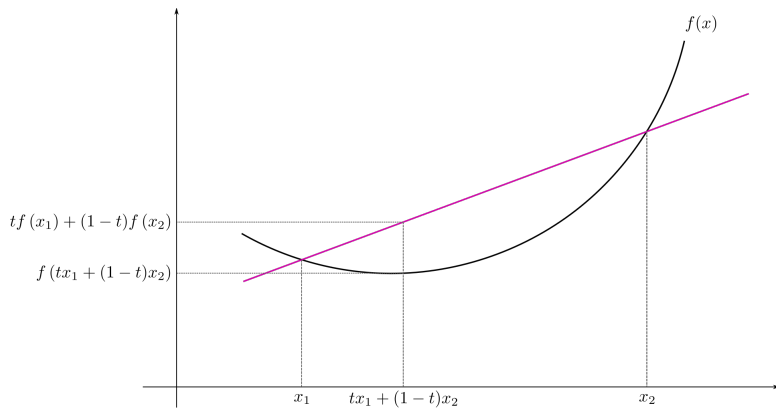
A function $f : D \rightarrow \mathbb{R}$ on convex D is **convex** if

$$f(tx_1 + (1 - t)x_2) \leq tf(x_1) + (1 - t)f(x_2)$$

(and **concave** if $\geq$).

For the strict version replace $\leq$ by $<$ for $t \in (0, 1)$ and $x_1 \neq x_2$.

Convex function



Quasi-convex/concave functions

f is quasi-convex if

$$f(tx_1 + (1 - t)x_2) \leq \max\{f(x_1), f(x_2)\}.$$

Quasi-concave if $\geq \min\{f(x_1), f(x_2)\}$.

Facts: convex $\Rightarrow$ quasi-convex; concave $\Rightarrow$ quasi-concave.

Any strictly increasing function is both quasi-convex and quasi-concave.

Economic examples:

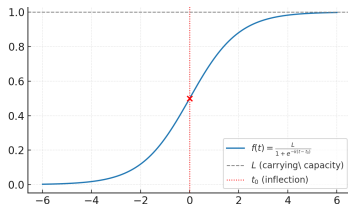
- **Indifference curves:** If a consumer is indifferent between two bundles, any convex combination is at least as good as one of them. This gives quasi-convex preferences.
- **Profits in price:** As a function of price, profits often rise, then fall (too cheap or too expensive reduces profit). This bell-shaped behavior is quasi-concave.

Economic example: logistic growth (adoption curves)

The logistic function with $L > 0$, $k > 0$, $t_0 \in \mathbb{R}$:

$$f(t) = \frac{L}{1 + e^{-k(t-t_0)}}$$

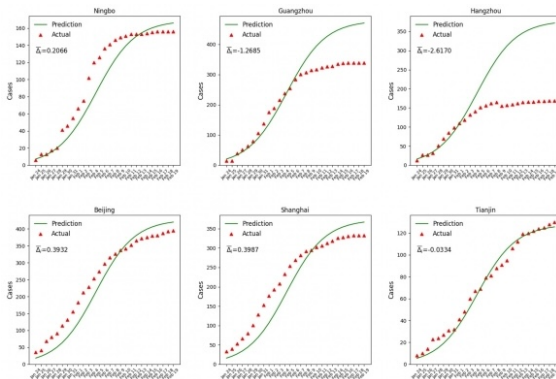
models **technology/product adoption** or **viral spread**. Early exponential growth, mid-phase inflection at t_0 , saturation near L (market size).

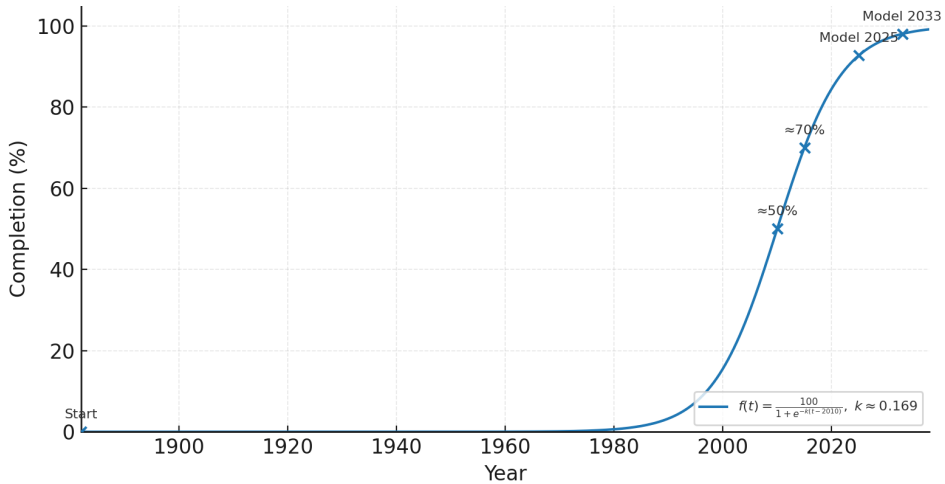


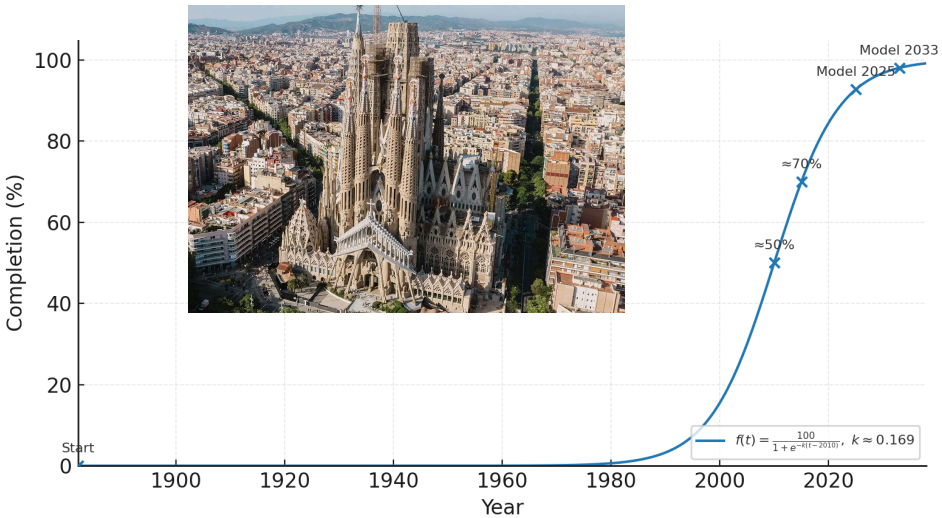
Interpretation: growth accelerates, then slows as saturation/frictions (capacity, regulation, budgets) bind. Same shape often appears in **learning curves** (accuracy vs. data).

Epidemics and beyond

Logistic curves fit epidemic growth, but also market penetration, platform adoption, content diffusion. Check the following epidemic curves over time of the COVID-19 in China







Examples of functions

1. **Monomials:** $f(x) = ax^k$, $a \in \mathbb{R}$.
2. **Polynomials:** finite sums of monomials.
3. **Rational:** ratios of polynomials.
4. **Exponential:** $f(x) = a^x$, $a > 0$.
5. **Trigonometric:** $\sin x$, $\cos x$, $\dots$
6. **Linear:** $f(x) = ax + b$ (slope a , intercept b ; also $y - f(x_0) = a(x - x_0)$).
7. **Natural exponential:** $f(x) = e^x$ ($e^x = \lim_n (1 + \frac{x}{n})^n$).
8. **Natural log:** $f(x) = \log x$ ($e^{\log x} = x$, $\log(e^x) = x$)

Economic examples (often based on the above fundamental examples):

- **Demand curves:** typically decreasing in price (law of demand).
- **Cost curves:** often convex, since producing extra units gets more expensive at higher scale.
- **Utility functions:** often concave, to capture *diminishing marginal utility* (each extra unit of a good adds less satisfaction).
- **Log-sum-exp:** arises in discrete choice models; connects probabilities to utilities.

Exponential functions

$$f(x) = 1^x$$

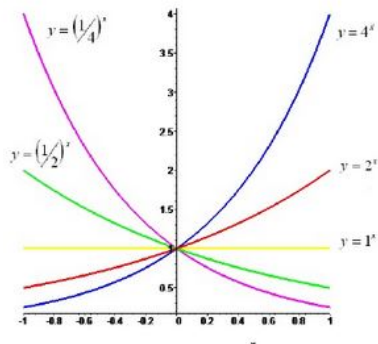
Horizontal line with y-intercept 1

$$f(x) = a^x, \quad 0 < a < 1$$

Exponential decay

$$f(x) = a^x, \quad a > 1$$

Exponential growth



Chapter 3: Differentiation

Differentiation measures sensitivity: how $f(x)$ changes with small changes in x

$$f(x + \Delta x) \approx f(x) + f'(x)\Delta x \quad \text{or, equivalently,} \quad \frac{f(x + \Delta x) - f(x)}{\Delta x} \approx f'(x).$$

Examples: price $\rightarrow$ demand sensitivity, ad spend $\rightarrow$ clicks, carbon tax $\rightarrow$ emissions, risk $\rightarrow$ portfolio value.

Calculus also underpins extrema, monotonicity, convexity — and the local behavior behind optimization and ML.

Read Ch. 4 Werner-Sotkov; Simon-Blume Chs. 2-5.

Exercises: 4.6(c), 4.10, 4.14, 4.15(a)-(c), 4.16(d), 4.17(b) (Werner-Sotkov), 5.5 (b)-(c)-(e) (Simon-Blume)

Functional Limits and Continuity

Limit of a function

Let $f : D \rightarrow \mathbb{R}$ and $x_0 \in \mathbb{R}$. We write $L = \lim_{x \rightarrow x_0} f(x)$ if

$$x_n \rightarrow x_0 \implies y_n := f(x_n) \rightarrow L.$$

Remark: x_0 need not be in D (removable/essential discontinuities matter in econ too).

Example: $\lim_{x \rightarrow 0} x \sin \frac{1}{x} = 0$ since $0 \leq |x \sin \frac{1}{x}| \leq |x|$.

Algebra of limits

If $\lim_{x \rightarrow x_0} f(x) = y_1$ and $\lim_{x \rightarrow x_0} g(x) = y_2$, then

$$\lim_{x \rightarrow x_0} (f(x) + g(x)) = y_1 + y_2, \quad \lim_{x \rightarrow x_0} (f(x)g(x)) = y_1 y_2, \quad \lim_{x \rightarrow x_0} \frac{f(x)}{g(x)} = \frac{y_1}{y_2} \text{ if } y_2 \neq 0.$$

Continuity

Definition

f is **continuous** at $x_0 \in D$ if $\lim_{x \rightarrow x_0} f(x) = f(x_0)$. If f is continuous at every point of $B \subset D$, we say f is continuous on B .

Theorem: If f, g are continuous at x_0 then so are $f + g$ and fg . If moreover, $g(x) \neq 0$ around x_0 , then $\frac{f}{g}$ is continuous at x_0 .

Heine-Borel fact (1D): continuous $f : [a, b] \rightarrow \mathbb{R}$ is bounded.

Non-example

Consider the function $f(x) = \frac{|x|}{x}$ for $x \neq 0$ and $f(0) = 1$. This function is not continuous. Why?

Composition and inverse

Two important results

(composition) If f, g are continuous, so is $h(x) = g(f(x))$ (we also write $h = g \circ f$); moreover

$$\lim_{x \rightarrow x_0} g(f(x)) = g\left(\lim_{x \rightarrow x_0} f(x)\right).$$

(inverse) If $f : [a, b] \rightarrow \mathbb{R}$ is strictly monotone and continuous with $f(a) = c, f(b) = d$, then the inverse $f^{-1} : [c, d] \rightarrow [a, b]$ is also strictly monotone and continuous.

Economic example: p = price of a good, q = quantity sold. The *demand function* $q(p)$ gives the quantity q consumers will buy at price p . Equivalently, the *inverse demand function* $p(q)$ gives the price at which q units can be sold. Revenue as a function of quantity is:

$$R(q) = p(q) \cdot q.$$



Differentiation

Derivative

Let $f : (a, b) \rightarrow \mathbb{R}$. f is differentiable at x if

$$\lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

exists. Denote this limit by $f'(x)$.

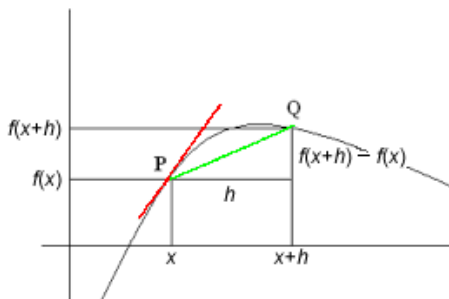
Fact: differentiable $\Rightarrow$ continuous.

The converse does not hold: $|x|$ is continuous but not differentiable at 0 (why?).

Geometric interpretation

If f is differentiable at x_0 , the tangent line to the graph of f at $(x_0, f(x_0))$ is

$$y = f(x_0) + f'(x_0)(x - x_0).$$



slope of **secant** through P, Q : $\frac{f(x+h) - f(x)}{h}$; slope of **tangent** at P : $f'(x)$

A “marginal” is a derivative

In economics, a **marginal** quantity measures how much something changes when we increase an input by a small amount. Mathematically, we often approximate

$$f(x + 1) - f(x) \approx f'(x),$$

when the change is 1 unit.

Examples:

- **Marginal revenue:** $\frac{d}{dq}[p(q) q]$ at the current sales level q . The derivative tells you how much extra revenue you gain (or lose) by selling one more unit.
- **Click-through sensitivity:** $\frac{d \text{CTR}}{d(\text{ad spend})}$ at the budget currently in use. Here CTR (click-through rate) is a function of ad spending; the derivative measures how much CTR improves with an extra euro of spending.

If the change is Δx rather than 1: $f(x + \Delta x) - f(x) \approx f'(x) \Delta x$.

In differential notation: $df \approx f'(x) dx$

Derivatives of elementary functions

1. $c' = 0$
2. $(x^n)' = nx^{n-1}$ for $n \in \mathbb{N}$
3. $(x^\alpha)' = \alpha x^{\alpha-1}$ for $\alpha \in \mathbb{R}, x > 0$
4. $(\log x)' = \frac{1}{x}, x > 0$
5. $(\sin x)' = \cos x$
6. $(\cos x)' = -\sin x$
7. $(e^x)' = e^x$
8. $(a^x)' = a^x \log a, a > 0$

Proof of 8: $(a^x)' = (e^{x \log a})' = a^x \log a$ (chain rule)

Product/quotient rules

If f, g are differentiable:

- $(f + g)' = f' + g'$
- $(fg)' = f'g + g'f$ [Leibniz rule]
- $\left(\frac{f}{g}\right)' = \frac{f'g - g'f}{g^2}$ where $g \neq 0$ near the point.

Example (revenue): $R(q) = p(q)q$, then $R'(q) = p'(q)q + p(q)$.

Chain rule

If $h = g \circ f$ and f, g are differentiable at the relevant points, then

$$h'(x) = g'(f(x)) f'(x).$$

Example (cost of production):

$$C(x) = 4 + \log(x + 1) + \sqrt{3x + 1} \quad \Rightarrow \quad C'(x) = \frac{1}{x + 1} + \frac{3}{2\sqrt{3x + 1}}.$$

Example (choice models): In discrete choice models, the logistic function $\sigma(x) = \frac{1}{1+e^{-x}}$ maps a score x (reflecting how attractive an option is) to the probability of choosing it. Its derivative $\sigma'(x) = \sigma(x)[1 - \sigma(x)]$ measures how sensitive the choice probability is to changes in x .

Differentiating $u(x)^{v(x)}$

Let $u(x) > 0$. For $h(x) = u(x)^{v(x)}$, taking logs gives:

$$\log h = v \log u \quad \Rightarrow \quad \frac{h'}{h} = v' \log u + v \frac{u'}{u}.$$

Hence

$$h'(x) = u(x)^{v(x)} \left(v'(x) \log u(x) + v(x) \frac{u'(x)}{u(x)} \right)$$

Econ example: A startup's valuation is $V(t) = V_0[1 + g(t)]^{T(t)}$, where $g(t)$ is the projected annual growth rate and $T(t)$ the number of years that growth lasts. Both $g(t)$ and $T(t)$ vary with t .

Some of the contributors (in form of *caganers*)



Newton



Leibniz



Euler



Lagrange

The inverse function theorem (1D version)

Theorem:

If f is strictly monotone, continuous, and differentiable at x with $f'(x) \neq 0$, then f^{-1} is differentiable at $y = f(x)$ and

$$(f^{-1})'(y) = \frac{1}{f'(x)} = \frac{1}{f'(f^{-1}(y))}.$$

Since $f^{-1}(f(x)) = x$, chain rule gives $(f^{-1})'(f(x))f'(x) = 1$.

Example: $f(x) = e^x \Rightarrow (\log)'(y) = 1/y$.

Higher-order derivatives: If f' is differentiable, its derivative is called the **second derivative** of f and denoted f'' . Similarly, we can define **higher-order derivatives**, and $f^{(n)}$ is called the n th derivative of f .

If $f^{(n)}$ is continuous, we say that f is n times **continuously differentiable** or **C^n** for short.

L'Hôpital's rule

Theorem: Let f, g be C^1 on (a, b) with $g' \neq 0$ on (a, b) . If

$$\lim_{x \rightarrow x_0} f(x) = \lim_{x \rightarrow x_0} g(x) = 0 \quad \text{or} \quad \pm \infty,$$

then

$$\lim_{x \rightarrow x_0} \frac{f(x)}{g(x)} = \lim_{x \rightarrow x_0} \frac{f'(x)}{g'(x)} \quad (\text{if the RHS limit exists}).$$

Proof idea uses the Mean Value Theorem (slide 57).

Example:

$$\begin{aligned} \lim_{x \rightarrow 0} (1+x)^{1/x} &= \lim_{x \rightarrow 0} \exp\left(\frac{1}{x} \log(1+x)\right) \\ &= \exp\left(\lim_{x \rightarrow 0} \frac{\log(1+x)}{x}\right) \\ &= \exp\left(\lim_{x \rightarrow 0} \frac{1}{1+x}\right) = e. \end{aligned}$$

Optimization in 1D

Monotonicity via derivatives

If f is differentiable on (a, b) :

1. f increasing on $[a, b] \iff f'(x) \geq 0$ on (a, b) .
2. f decreasing on $[a, b] \iff f'(x) \leq 0$ on (a, b) .
3. f constant on $[a, b] \iff f'(x) = 0$ on (a, b) .
4. $f'(x) > 0$ on $(a, b) \Rightarrow f$ strictly increasing on $[a, b]$.
5. $f'(x) < 0$ on $(a, b) \Rightarrow f$ strictly decreasing on $[a, b]$.

Converse of 4-5 can fail at isolated points (e.g., x^3).

Example: $f(x) = \frac{x^3}{3} + 2x^2 + 3x + 1$; $f'(x) = (x + 1)(x + 3)$ determines monotonicity intervals and local extrema.

Optimal points: first-order conditions

Local optima

Let $f : D \rightarrow \mathbb{R}$. A point $x_0 \in D$ is a **local maximum** if $\exists (a, b) \subset D$ with $x_0 \in (a, b)$ and $f(x) \leq f(x_0)$ for all $x \in (a, b)$. Similarly for **local min** with $\geq$.

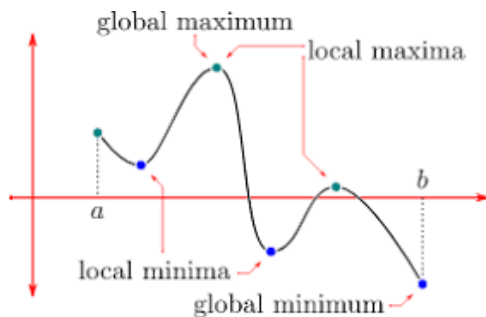
If f has a local optimum at interior x and is differentiable, then $f'(x) = 0$.

To see why this is true, stare at $f(x_0 + h) - f(x_0) \approx f'(x_0)h$.

This condition is **necessary**, not sufficient; see e.g. x^3 .

If x is stationary ($f'(x) = 0$) and f is strictly increasing to the left and strictly decreasing to the right, then x is a local max (reverse for min).

Optimal points



Example: Timing a sale (present value)

Suppose an assets market value is $V(t)$ at time t . At a constant annual interest rate R , the **present value** of selling at time t is

$$P(t) = V(t) e^{-Rt},$$

where e^{-Rt} discounts future cash flows.

Goal: Choose t to maximize $P(t)$.

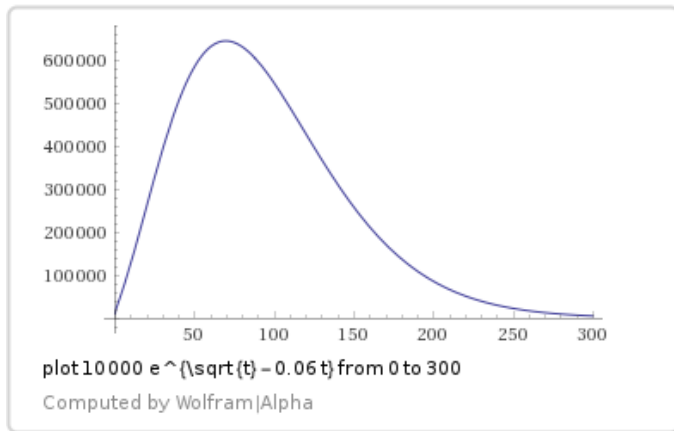
FOC (set derivative to zero):

$$P'(t) = V'(t)e^{-Rt} - R V(t)e^{-Rt} = 0 \quad \Rightarrow \quad \frac{V'(t)}{V(t)} = R.$$

Interpretation: Sell when the assets *instantaneous percentage growth rate* (matches) the *interest rate you could earn in the bank*.

Example: If $V(t) = 10000 e^{\sqrt{t}}$ and $R = 6\%$, then $P(t) = 10000 e^{\sqrt{t} - 0.06t}$, which is maximized near $t \approx 69.44$.

Plot of the present value $P(t)$



To certify a **global** optimum on an interval, compare values at boundaries and critical points. Here $P(0) = 10\,000$, $\lim_{t \rightarrow \infty} P(t) = 0$.

Second-order conditions

Test: If $f'(x_0) = 0$ and $f''(x_0) < 0$ (> 0), then x_0 is a strict local max (min).

Indeed,
$$f(x_0 + h) \approx f(x_0) + f'(x_0)h + \frac{1}{2}f''(x_0)h^2 = f(x_0) + \frac{1}{2}f''(x_0)h^2.$$

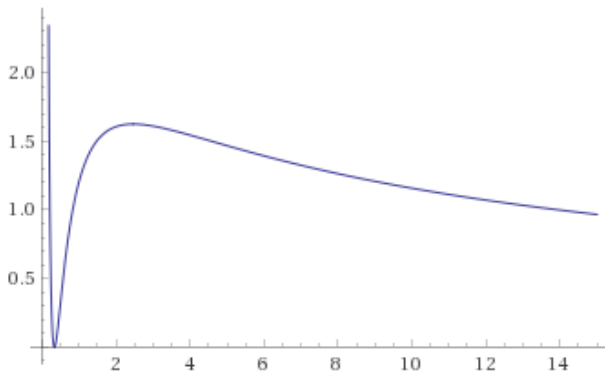
Example: For $f(x) = \frac{\log^2(3x)}{x}$ ($x > 0$):

$$f'(x) = \frac{(2 - \log 3x) \log 3x}{x^2}, \quad f''(x) = \frac{2(1 - 3 \log 3x + \log^2 3x)}{x^3}.$$

Critical points at $x_1 = \frac{1}{3}$ (min), $x_2 = \frac{e^2}{3}$ (max).

Question: What if $f''(x_0) = 0$?

Plot of $f(x) = \frac{\log^2 3x}{x}$, $x > 0$



plot $\ln^2(3x)/x$ from 0 to 15 | Computed by Wolfram|Alpha

$\lim_{x \rightarrow \infty} f(x) = 0$, $\lim_{x \rightarrow 0^+} f(x) = \infty$, $f(x_1) = 0$, $f(x_2) > 0 \Rightarrow x_1$ global min, no global max.

Convexity and concavity

If f is twice differentiable on (a, b) :

1. f convex on $[a, b] \Leftrightarrow f''(x) \geq 0$ for all $x \in (a, b)$.
2. f concave on $[a, b] \Leftrightarrow f''(x) \leq 0$ for all $x \in (a, b)$.
3. $f'' > 0$ (< 0) $\Rightarrow f$ strictly convex (concave).

Theorem: If f and g are convex (or concave), then $f \circ g$ is convex (or concave).

Example: $f(x) = e^{x^2}$ is strictly convex on $[0, \infty)$.

Economics:

- **Cost functions:** usually convex — producing extra units becomes progressively more expensive (e.g. overtime wages, machine wear).
- **Utility functions:** often concave — reflects diminishing marginal utility (the first slice of pizza gives more satisfaction than the tenth).
- **Machine learning:** convex losses (e.g. squared error, logistic loss + ℓ_2 regularization) ensure optimization problems are tractable and stable.

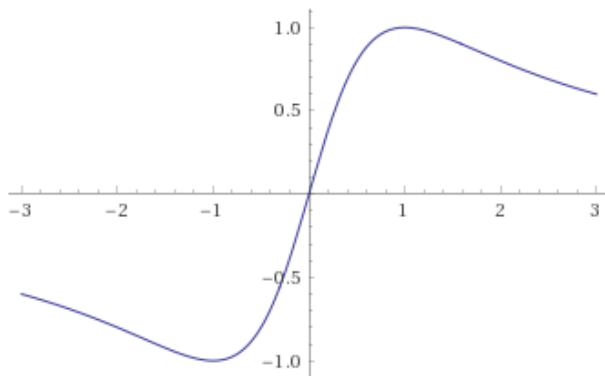
Example of changing curvature

Let $f(x) = \frac{2x}{x^2 + 1}$. Then

$$f'(x) = \frac{2(1 - x^2)}{(x^2 + 1)^2}, \quad f''(x) = \frac{4(x^3 - 3x)}{(x^2 + 1)^3}.$$

So f is strictly convex on $[-\sqrt{3}, 0] \cup [\sqrt{3}, \infty)$ and strictly concave on $(-\infty, -\sqrt{3}] \cup [0, \sqrt{3}]$.

Plot of $f(x) = \frac{2x}{x^2 + 1}$



plot $2x/(x^2+1)$ from -3 to 3 | Computed by Wolfram|Alpha

Inflexion points at $x \in \{-\sqrt{3}, 0, \sqrt{3}\}$. Global max at $x = 1$, min at $x = -1$, and $\lim_{|x| \rightarrow \infty} f(x) = 0$.

Inflexion points

If f is C^n on (a, b) , an **inflexion point** at x_0 occurs when curvature changes sign. A sufficient test:

$$f''(x_0) = \dots = f^{(n-1)}(x_0) = 0, \quad f^{(n)}(x_0) \neq 0, \quad n \text{ odd.}$$

Example: $f(x) = x^3$ has $f''(0) = 0$, $f^{(3)}(0) = 6 \neq 0 \Rightarrow$ inflexion at 0.

Taylor series expansion

Mean-value theorem

Mean Value Theorem

If f is continuous on $[a, b]$ and differentiable on (a, b) , then $\exists c \in (a, b)$ such that

$$f'(c) = \frac{f(b) - f(a)}{b - a} \quad (\text{equivalently } f(b) = f(a) + f'(c)(b - a)).$$

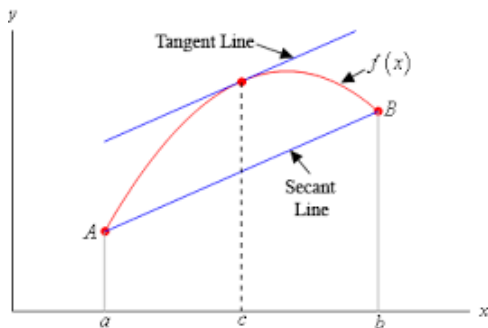
Proof idea: consider $h(x) = (f(b) - f(a))x - (b - a)f(x)$; since $h(a) = h(b)$, h has an interior extremum.

Interpretation: the instantaneous slope matches the secant slope at some point.

Geometry of the mean-value theorem

Recall: $\exists c \in (a, b)$ s.t.

$$f'(c) = \frac{f(b) - f(a)}{b - a}.$$



Taylor formula

Theorem (Taylor's formula of order n): Let f be $n + 1$ times differentiable on (a, b) , and let $x_0 \in (a, b)$ given. Then, for any $x \in (a, b)$, if $n = 1$,

$$f(x) = f(x_0) + f'(x_0)(x - x_0) + \text{error}.$$

If $n \geq 2$,

$$f(x) = f(x_0) + f'(x_0)(x - x_0) + \frac{f''(x_0)}{2!}(x - x_0)^2 + \frac{f^{(n)}(x_0)}{n!}(x - x_0)^n + \text{error}.$$

The error is small if x is close to x_0 , therefore, this theorem says that differentiable functions may be **locally approximated by a polynomial** called the **Taylor polynomial**.

Economics/data: local linear/quadratic approximations underlie elasticity (log-linearization), Delta method, and second-order price responses.

Taylor remainder and a handy expansion

Remainder (Lagrange form)

$$R_n^y(x) = \frac{f^{(n+1)}(y)}{(n+1)!} (x - x_0)^{n+1} \quad \text{for some } y \text{ between } x_0 \text{ and } x.$$

If $|f^{(n+1)}| \leq M$ near x_0 , then $|R_n^y(x)| \leq \frac{M}{(n+1)!} |x - x_0|^{n+1}$.

Example (log around 1): With $x_0 = 1$,

$$\log x \approx (x - 1) - \frac{1}{2}(x - 1)^2, \quad R_2^y(x) = \frac{1}{3y^3}(x - 1)^3.$$

Finance quick check: For small r , $\log(1 + r) \approx r - \frac{r^2}{2} \Rightarrow \text{EAR} \approx \text{APR} - \frac{1}{2}\text{APR}^2$ (small correction).

Short comment about small-o notation

Writing $|R_n^y(x)| \leq \frac{M}{(n+1)!}|x-x_0|^{n+1}$ we conclude $\frac{R_n^y(x)}{|x-x_0|^n} \leq \frac{M}{(n+1)!}|x-x_0|$ and so

$$\lim_{x \rightarrow x_0} \frac{R_n^y(x)}{|x-x_0|^n} = 0.$$

This we write as $R_n^y(x) = o(|x-x_0|^n)$.

In particular: If f is twice differentiable:

$$f(x+h) = f(x) + f'(x)h + o(|h|).$$

The generalization of this last expression to multivariate functions will provide powerful insights into their optimization.